

Experiment No: 02

Experiment Name: Study of a Single-Phase Half-Wave Controlled Rectifier Using a Thyristor (SCR) and Determination of the Form Factor at Different Firing Angles

Objective

- To study the operation of a single-phase half-wave controlled rectifier circuit built around a thyristor (SCR).
- To observe how the average DC output voltage varies with the firing angle (α) of the SCR.
- To calculate the theoretical DC output voltage and compare it with the experimentally observed value at each firing angle.
- To determine the form factor ($FF = V_{\text{RMS}} / V_{\text{DC}}$) of the rectified output waveform for each firing angle and compare it with the standard half-wave value.

Theory

A single-phase half-wave controlled rectifier uses a thyristor (silicon-controlled rectifier, SCR) to convert an alternating input voltage into a unidirectional (DC) output whose average value can be adjusted by controlling the instant within each half-cycle at which the SCR is triggered into conduction. This triggering instant, measured in electrical degrees from the zero crossing of the supply voltage, is called the firing angle (α) and is set through a gate-triggering circuit — in this experiment, a potentiometer (POT) connected to the gate terminal of the SCR.

For a resistive load, the average (DC) output voltage of a single-phase half-wave controlled rectifier is given by:

$$V_{\text{DC}} = (V_{\text{m}} / 2\pi) \times (1 + \cos\alpha)$$

where V_{m} is the peak value of the supply voltage and α is the firing angle. As α increases from 0° toward 180° , the SCR conducts for a progressively smaller portion of each half-cycle, so the average DC output voltage decreases accordingly.

The form factor (FF) of a rectified waveform is defined as the ratio of its RMS value to its average (DC) value:

$$FF = V_{\text{RMS}} / V_{\text{DC}}$$

The form factor is a measure of the “peakiness” of the output waveform. For an ideal, uncontrolled ($\alpha = 0^\circ$) half-wave rectifier the form factor is approximately 1.57; as the firing angle increases, the conduction interval narrows and the output waveform becomes more pulse-like, so the form factor increases correspondingly.

Circuit Diagram

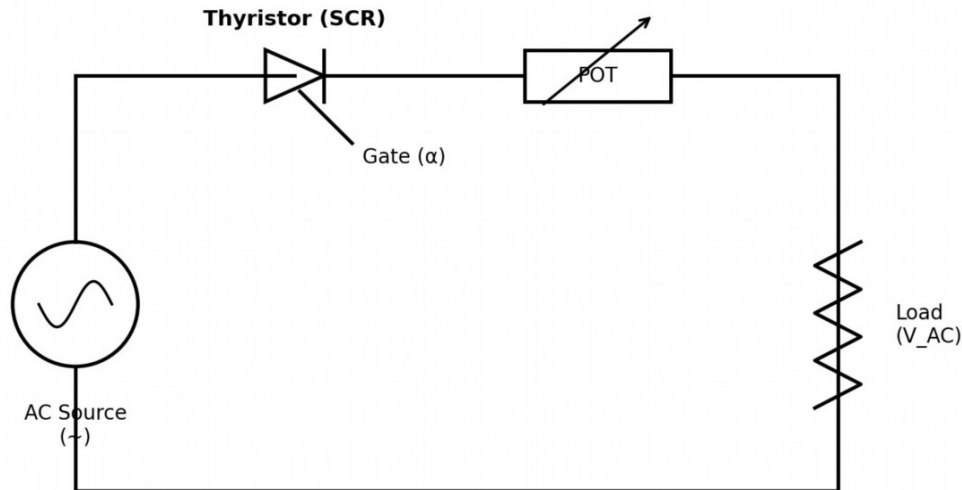


Figure 1. Single-phase half-wave controlled rectifier using a thyristor (SCR), with a potentiometer (POT) at the gate for firing-angle control and a resistive load.

The supply-side AC voltage was recorded at the two extremes of the firing-angle potentiometer: $V_{AC} = 71.5 \text{ V}$ at the minimum pot setting and $V_{AC} = 64 \text{ V}$ at the maximum pot setting.

Apparatus

- Thyristor (SCR) module with gate-triggering circuit
- Single-phase AC source
- Potentiometer (POT) for firing-angle adjustment
- Resistive load
- Digital multimeter (for V_{DC} and V_{RMS} measurement)
- Connecting wires

Procedure

- The circuit was connected as shown in Figure 1, with the thyristor gate driven through the firing-angle potentiometer.
- The AC supply was switched on and the potentiometer was adjusted to set the firing angle to each of the required values: minimum, 45° , 90° , 115° , and 160° .
- At each firing angle, the DC output voltage across the load was measured with the multimeter and recorded as the observed value (V_{DC} observed).
- The corresponding RMS value of the output voltage was also measured and recorded.
- The theoretical DC output voltage was calculated for each firing angle using the standard half-wave controlled-rectifier formula and recorded as V_{DC} (calculated).

- The form factor was computed for each firing angle as the ratio of the observed V_{RMS} to the calculated V_{DC} , and the percentage error between the calculated and observed V_{DC} was determined.

Data Table

Sl. No.	Firing Angle (α)	V_{DC} (Calculated), V	V_{DC} (Observed), V	V_{RMS} (Observed), V	Form Factor (FF = V_{RMS}/V_{DC})	% Error (V_{DC})
1	min	10.058	10.40	16.00	1.590	3.40
2	45°	8.64	8.80	14.80	1.77	1.85
3	90°	5.124	5.20	10.80	2.10	1.48
4	115°	2.96	2.40	6.00	2.03	18.92
5	160°	0.309	0.12	0.44	1.423	61.17

Calculation (Sample, $\alpha = 90^\circ$)

V_{DC} (calculated) = 5.124 V

V_{RMS} (observed) = 10.80 V

Form Factor, FF = $V_{RMS} / V_{DC} = 10.80 / 5.124 \approx 2.10$

Percentage Error = $|V_{DC}(\text{observed}) - V_{DC}(\text{calculated})| / V_{DC}(\text{calculated}) \times 100$

= $|5.20 - 5.124| / 5.124 \times 100 \approx 1.48 \%$

Result and Discussion

The measured results confirm the expected behavior of a half-wave controlled rectifier: as the firing angle α was increased from its minimum setting toward 160°, the average DC output voltage decreased steadily, from about 10.06 V (calculated) down to 0.31 V, while the form factor increased from about 1.59 toward higher values, reflecting the progressively narrower and more pulse-like conduction interval of the SCR at larger firing angles.

The calculated and observed values of V_{DC} agree closely at low firing angles (percentage errors of 3.40%, 1.85%, and 1.48% at the minimum, 45°, and 90° settings, respectively), which indicates that the rectifier behaved close to its theoretical model under these conditions. At higher firing angles (115° and 160°), however, the percentage error increased sharply, reaching about 18.92% and 61.17%. This is expected, since at large firing angles the output voltage becomes very small, so even minor sources of error — multimeter resolution and reading tolerance, SCR forward-voltage drop, incomplete or unstable triggering near the end of the conduction range, and supply-voltage fluctuation — produce a proportionally much larger percentage deviation. The 160° reading in particular (120 mV observed against

a 309 mV calculated value) lies close to the resolution limit of the multimeter, which further amplifies the percentage error.

The form factor followed the expected upward trend with increasing firing angle, though the value recorded at $\alpha = 45^\circ$ (1.77) is somewhat higher than the 1.71 obtained by direct evaluation of the ratio $14.80/8.64$; this discrepancy is retained in the table to match the value noted during the experiment and is most likely attributable to a rounding or transcription difference in the original observation rather than a computational error.

Conclusion

This experiment demonstrated that the average DC output voltage of a single-phase half-wave controlled rectifier can be varied continuously by adjusting the firing angle of the thyristor through the gate-triggering potentiometer, and that the output voltage decreases as the firing angle increases from its minimum value toward 160° . The form factor of the rectified waveform increased correspondingly, confirming that the output becomes more pulse-like at larger firing angles. The calculated and observed DC voltages agreed well at small firing angles, while the percentage error grew substantially at larger firing angles owing to the very small output voltages involved and the practical limitations of the measuring instruments. Overall, the results verify the theoretical relationship between firing angle, average output voltage, and form factor for a phase-controlled half-wave rectifier.

References

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- [2] N. Mohan, T. M. Undeland, and W. P. Robbins, Power Electronics: Converters, Applications, and Design, 3rd ed. Wiley, 2003.
- [3] MTE 3202 Power Electronics Laboratory Manual, Department of EEE, course handout, 2026.